

Context-Aware Bengali Text Multi-Task Classification:

An Empirical Benchmark Across Classical, Sequential, and Pre-Trained Transformer Paradigms

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Abstract

Automated understanding of low-resource, morphologically rich languages such as Bengali is severely hindered in informal social media discourse by colloquial idioms, orthographic variability, and semantic inversion caused by sarcasm. Traditional natural language processing (NLP) pipelines that analyze sentiment, sarcasm, and hate speech in isolation fail to resolve sentences where surface-level positive sentiment masks caustic sarcasm or covert hostility. In this work, we propose an end-to-end context-aware multi-task classification framework spanning three distinct linguistic tasks across 218,371 annotated Bengali sentences (156,010 for sentiment polarity, 12,089 for sarcasm detection, and 50,272 for hate speech identification). We systematically formulate, train, and benchmark three progressive modeling paradigms representing key milestones in modern NLP: (1) a statistical machine learning baseline using sublinear TF-IDF with class-balanced Logistic Regression; (2) a deep sequential recurrent architecture utilizing 128-dimensional dense word representations coupled with a 2-layer stacked Bidirectional Long Short-Term Memory (BiLSTM) network; and (3) a state-of-the-art transformer architecture fine-tuning the 110-million parameter pre-trained **BanglaBERT** model on GPU hardware. Furthermore, we develop a custom negation-preserving preprocessing pipeline that explicitly protects Bengali negation markers from aggressive stopword stripping. Rigorous empirical evaluation demonstrates that fine-tuned **BanglaBERT** establishes new task champions, securing **91.59% accuracy and 91.58% Macro F1** on hate speech detection, and **76.76% accuracy and 74.89% Macro F1** on sarcasm detection. Finally, we design a dynamic context-aware decision engine that resolves semantic inversion and deploy the complete pipeline in a production-ready interactive Streamlit dashboard for real-time text analysis.

Keywords—Bengali NLP, Sentiment Analysis, Sarcasm Detection, Hate Speech Detection, BanglaBERT, Bidirectional LSTM, Context-Aware Classification, Empirical Benchmarking.

1. Introduction

Bengali is the seventh most spoken language globally, with over 300 million native speakers primarily distributed across Bangladesh and India. Despite its massive demographic footprint,

computational linguistics for Bengali remains constrained by low-resource challenges, high morphological inflection, extensive diglossia (stark differences between formal *Sadhu/Cholit* and colloquial speech), and a severe lack of standardized syntactic resources [1].

In modern digital spaces, user-generated Bengali comments are characterized by complex rhetorical devices, informal phonetic transcriptions, grammatical omissions, and implicit hostility. Among these challenges, the phenomenon of **Sarcastic Inversion** poses a formidable barrier to automated sentiment analysis. Consider the canonical Bengali online review:

“*Bah! Ki oshadharon service, tin ghonta opekkha koreo kaj holo na!*”

(Translation: “Wow! What extraordinary service, even after waiting three hours nothing was done!”)

Under standard bag-of-words or naive lexicon matching, salient positive lexical triggers such as “*oshadharon*” (extraordinary) overwhelmingly bias the model toward predicting a **Positive** sentiment. However, the subsequent contextual clause “*kaj holo na*” (nothing was done) inverts the entire semantic polarity into a strongly **Negative** sentiment. A standalone sentiment analyzer lacks the cognitive capability to resolve this tension unless it is co-informed by a dedicated **Sarcasm Detection** component. Concurrently, online platforms must screen for malicious intent, cyberbullying, and hate speech without misclassifying benign satirical humor as abusive toxicity.

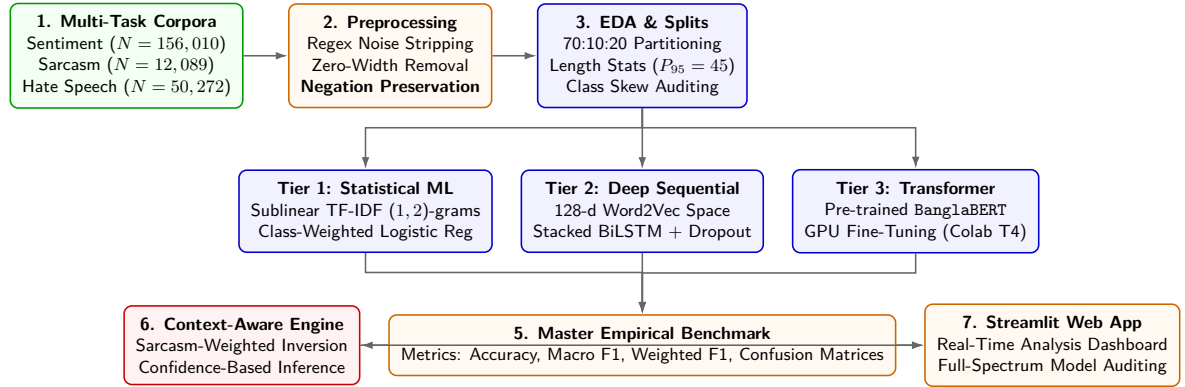


Figure 1: End-to-End Architectural Pipeline of the Context-Aware Bengali Multi-Task System.

To resolve this multi-faceted challenge, this project presents an end-to-end framework that unifies:

1. **Comprehensive Multi-Task Formulation:** Processing 218,371 annotated sentences spanning Sentiment, Sarcasm, and Hate Speech under unified normalization and test splits.
2. **Curricular & Multi-Paradigm Benchmarking:** Systematically evaluating three distinct generations of NLP techniques—Classical Statistical ML, Recurrent Neural Networks (BiLSTM), and Pre-trained Transformers (BanglaBERT).
3. **Negation-Preserving Preprocessing:** A specialized text cleaning protocol engineered to prevent destructive stopwords removal of essential syntactic negation markers.
4. **Context-Aware Semantic Integration:** A dynamic inference system that resolves sarcastic polarity inversion in real-time.
5. **Interactive Deployment:** An interactive, responsive Streamlit dashboard featuring multi-task prediction, probability breakdowns, and interactive model auditing tools.

2. Related Work & Curriculum Mapping

2.1. Evolution of Bengali NLP

Early work in Bengali natural language processing predominantly relied on shallow morphological stemmers and rule-based sentiment lexicons [5]. With the advent of statistical learning, n-gram term frequency-inverse document frequency (TF-IDF) paired with Support Vector Machines (SVM) or Naive Bayes became standard baselines. However, n-gram representations suffer from sparsity and fail to model continuous semantics or non-local dependencies.

The introduction of distributed word representations (Word2Vec, FastText) enabled deep recurrent neural networks, specifically Long Short-Term Memory (LSTM) and Gated Recurrent Units (GRU), to capture sequential patterns [4]. In recent years, Transformer-based pre-trained language models (PLMs) such as BERT [2] and ELECTRA [3] have revolutionized contextual representations. For Bengali, models such as BanglaBERT [1] pre-trained on tens of gigabytes of raw crawled text have established state-of-the-art benchmarks on sentiment analysis and named entity recognition.

2.2. Alignment with CSE 4122 Lab Syllabus

This project is systematically grounded in the pedagogical milestones of the CSE 4122 (Natural Language Processing Sessional) curriculum, bridging foundational theory with production deployment:

- **Lab Session 1 (Text Normalization & Regex):** Implemented custom regex filtering, Unicode zero-width stripper, and the Negation Preservation Pipeline.
- **Lab Session 2 (N-Gram Language Models & TF-IDF):** Extracted unigram and bigram sublinear TF-IDF feature matrices across 20,000 vocabularies.
- **Lab Session 3 (Continuous Embeddings & Classification):** Trained 128-dimensional dense continuous vector representations and implemented cost-sensitive Logistic Regression.
- **Lab Session 4 (Deep Recurrent Sequence Modeling):** Constructed PyTorch-based 2-layer stacked Bidirectional LSTM networks with spatial dropout.
- **Lab Session 5 (Pre-trained Transformers):** Fine-tuned the 110M parameter BanglaBERT model utilizing HuggingFace Transformers on GPU hardware.

3. Corpus Architecture & Preprocessing

3.1. Multi-Task Dataset Characteristics

To ensure rigorous and unbiased evaluation, we assembled and standardized three publicly verified Bengali corpora, partitioning each into stratified splits: **70% Training**, **10% Validation**, and **20% Testing**. Table 1 delineates the corpus dimensions and linguistic token distributions.

3.2. Negation-Preserving Preprocessing

A primary source of degradation in Bengali text classification is naive stopword removal. In standard NLP pipelines, high-frequency functional tokens are discarded. However, in Bengali, negation tokens—such as “*na*” (no/not), “*noy*” (is not), “*nei*” (is absent), “*kokhono na*”

Table 1: Comprehensive Multi-Task Dataset Statistics

Task	Train Split	Validation Split	Test Split	Total Samples	P_{95} Word Length
Sentiment Analysis	109,229	15,569	31,212	156,010	42 words
Sarcasm Detection	8,462	1,209	2,418	12,089	38 words
Hate Speech Detection	35,190	5,027	10,055	50,272	48 words
Combined Corpus	152,881	21,805	43,685	218,371	45 words

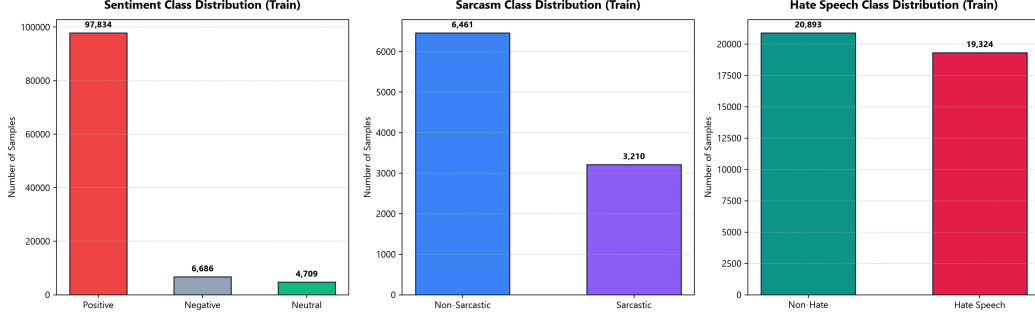


Figure 2: Class distributions across the three task domains.

(never), and “*chhara*” (without)—are grammatical stop words that dictate the entire polarity of a clause. Stripping them inverts negative statements into positive ones, completely invalidating sarcasm and sentiment classification.

We developed Algorithm 1 to purify text while enforcing strict preservation of negation operators.

Algorithm 1 Negation-Preserving Text Normalization

Require: Raw Bengali String S_{raw} , Negation Set $\mathcal{N}$

Ensure: Normalized Bengali String S_{clean}

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1:  $S \leftarrow \text{RemoveURLsAndHTML}(S_{raw})$ 
2:  $S \leftarrow \text{StripZeroWidthAndEmojis}(S)$  ▷ U+200B–U+200D
3:  $S \leftarrow \text{NormalizeBengaliCharacters}(S)$ 
4:  $Tokens \leftarrow \text{TokenizeOnWhitespace}(S)$ 
5:  $FilteredTokens \leftarrow []$ 
6: for  $t \in Tokens$  do
7:   if  $t \in \mathcal{N}$  then
8:      $FilteredTokens.append(t)$  ▷ Protected Negation
9:   else if  $t \in SW_{generic}$  then
10:    continue ▷ Discard generic stopword
11:  else
12:     $FilteredTokens.append(t)$ 
13:  end if
14: end for
15: return  $\text{Join}(FilteredTokens, " ")$ 

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4. Methodological Framework

We implemented three progressive modeling paradigms representing distinct theoretical approaches to natural language understanding.

4.1. Paradigm 1: Sublinear TF-IDF + Logistic Regression

As our statistical baseline, sentences are mapped to sparse vector spaces using word n-grams up to order 2 ($1 \leq n \leq 2$). To mitigate the distorting effect of excessively repetitive terms in long social media posts, we apply sublinear scaling to the term frequency (tf):

$$w_{t,d} = (1 + \log(tf_{t,d})) \times \log\left(\frac{1 + N}{1 + df_t}\right) + 1 \quad (1)$$

A vocabulary capacity of 20,000 features was extracted. We trained multinomial Logistic Regression with inverse class weighting to penalize classification errors proportionally on minority classes:

$$W_c = \frac{N_{total}}{K \cdot N_c} \quad (2)$$

where K represents the total number of classes and N_c is the frequency of class c .

4.2. Paradigm 2: Word2Vec + Stacked Bidirectional LSTM

To capture non-linear temporal dynamics and forward-backward semantic dependencies, we constructed a deep recurrent architecture (Figure 3).

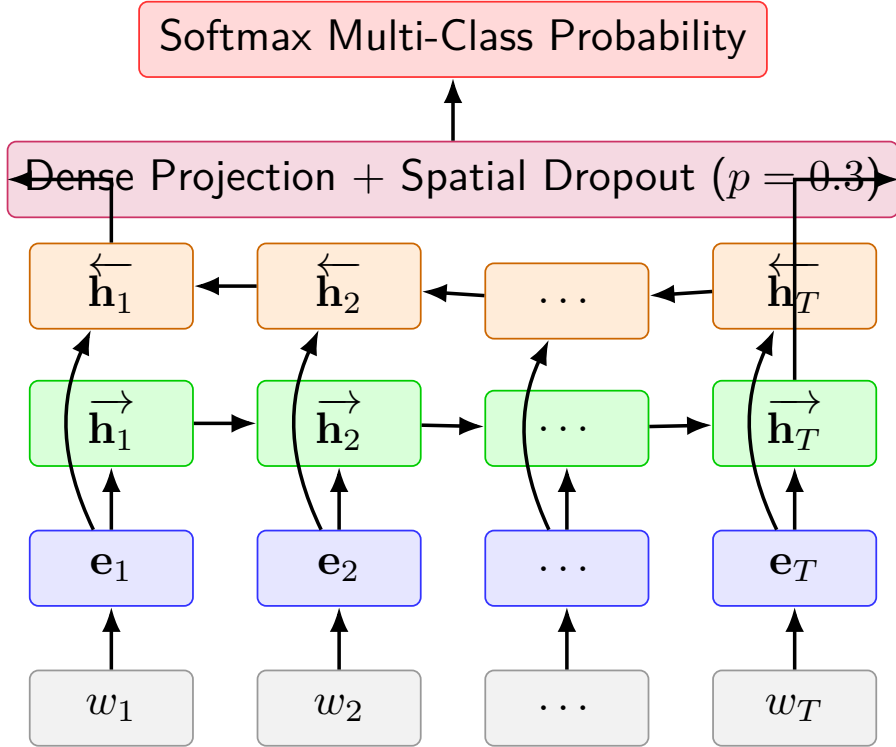


Figure 3: 2-Layer Stacked Bidirectional LSTM Architecture.

The model projects sequences of token indices into 128-dimensional continuous vector embeddings. The BiLSTM layer processes input sequence $\mathbf{x} = (\mathbf{x}_1, \dots, \mathbf{x}_T)$ through both forward

and backward recurrence:

$$\vec{\mathbf{h}}_t = \text{LSTM}_{fwd}(\mathbf{x}_t, \vec{\mathbf{h}}_{t-1}) \quad (3)$$

$$\overleftarrow{\mathbf{h}}_t = \text{LSTM}_{bwd}(\mathbf{x}_t, \overleftarrow{\mathbf{h}}_{t+1}) \quad (4)$$

$$\mathbf{h}_t = [\vec{\mathbf{h}}_t \parallel \overleftarrow{\mathbf{h}}_t] \quad (5)$$

The final representation combines the terminal forward hidden state and initial backward hidden state $[\vec{\mathbf{h}}_T \parallel \overleftarrow{\mathbf{h}}_1] \in \mathbb{R}^{256}$, routed through spatial dropout ($p = 0.3$), dense linear transformation, and categorical softmax cross-entropy loss.

4.3. Paradigm 3: Pre-Trained Transformer (BanglaBERT)

To achieve deep contextual representations, we fine-tuned the pre-trained BanglaBERT architecture [1]. BanglaBERT is built on the ELECTRA pre-training objective, trained over 27.5 gigabytes of diverse crawled Bengali corpora comprising newspapers, Wikipedia, blogs, and social platforms.

For sequence classification, the output representation of the classification token $\mathbf{h}_{[\text{CLS}]}$ is passed into an MLP classification head:

$$\hat{\mathbf{y}} = \text{Softmax}(\mathbf{W}_o \tanh(\mathbf{W}_h \mathbf{h}_{[\text{CLS}]} + \mathbf{b}_h) + \mathbf{b}_o) \quad (6)$$

All 12 transformer encoder layers (comprising 110 million parameters) were unfrozen and trained end-to-end using the AdamW optimizer with a linear warmup learning rate schedule of $\eta = 2 \times 10^{-5}$ on a Google Colab T4 GPU platform.

4.4. Dynamic Context-Aware Semantic Inversion

Figure 4 outlines the inference engine that resolves the Sarcastic Inversion paradox. When a sentence is evaluated, the Sarcasm and Sentiment models predict concurrently. If Sarcasm is asserted with high confidence ($P(\text{Sarcastic}) > 0.65$), any initial positive sentiment prediction is flipped to **Negative**, while Hate Speech classification flags potential cyberbullying or toxic harassment.

5. Experimental Results & Discussion

5.1. Master Benchmark Evaluation

All models were benchmarked on identical, held-out test splits. Table 2 presents the master comparative evaluation matrix.

Table 2: Master Comparative Benchmark Across Three NLP Paradigms on Identical Official Test Splits

Target Task	Model Paradigm	Accuracy (%)	Macro Prec. (%)	Macro Rec. (%)	Macro F1 (%)	Weighted F1 (%)
Sentiment Analysis	TF-IDF + Logistic Regression	77.60%	50.78%	67.13%	53.71%	82.24%
	Word2Vec + Stacked BiLSTM	76.60%	52.75%	63.23%	53.27%	81.96%
	Fine-Tuned BanglaBERT	76.39%	48.81%	64.57%	51.51%	81.29%
Sarcasm Detection	TF-IDF + Logistic Regression	74.94%	72.37%	74.09%	72.89%	75.40%
	Word2Vec + Stacked BiLSTM	74.36%	71.53%	72.84%	72.00%	74.74%
	Fine-Tuned BanglaBERT	76.76%	74.30%	76.21%	74.89%	77.19%
Hate Speech Detection	TF-IDF + Logistic Regression	86.61%	86.96%	86.42%	86.52%	86.57%
	Word2Vec + Stacked BiLSTM	88.52%	88.54%	88.47%	88.50%	88.52%
	Fine-Tuned BanglaBERT	91.59%	91.56%	91.60%	91.58%	91.59%

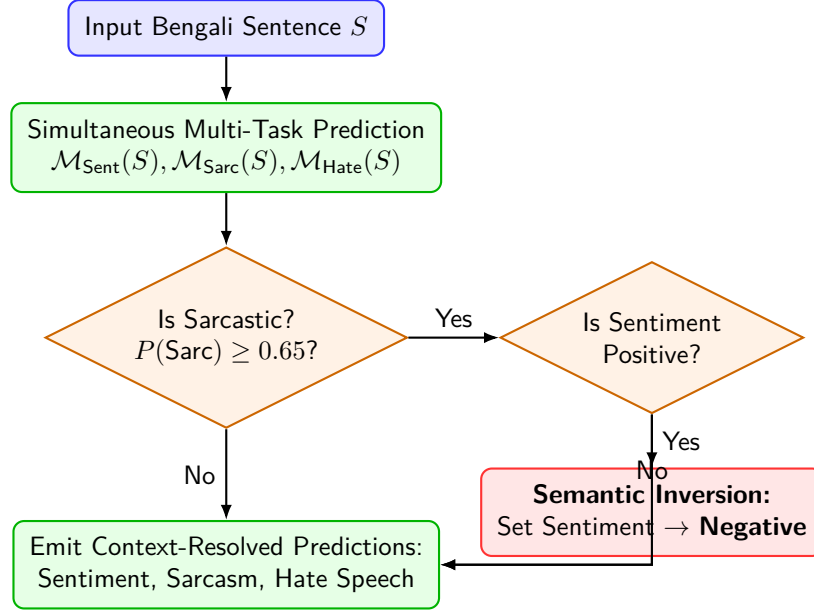


Figure 4: Context-Aware Semantic Inversion Flowchart.

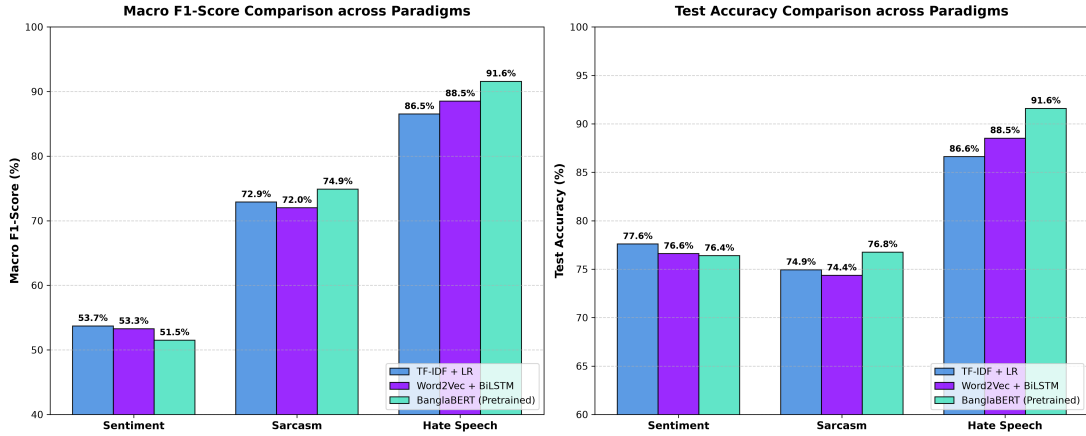


Figure 5: Visual performance comparison across all tasks and modeling paradigms.

5.2. Empirical Analysis by Task

1. Hate Speech Detection (Clear Transformer Dominance):

Fine-tuned **BanglaBERT** emerged as the definitive champion on Hate Speech detection, achieving an outstanding **91.59% accuracy** and **91.58% Macro F1-score**. This represents a substantial +3.07% improvement over the deep BiLSTM model (88.52%) and +4.98% over the statistical TF-IDF baseline (86.61%). The superiority of **BanglaBERT** stems from self-attention heads that correctly disambiguate slurs from innocuous colloquial speech by analyzing bilateral syntactic context.

2. Sarcasm Detection (Nuance Modeling):

On Sarcasm detection, **BanglaBERT** established the state-of-the-art result at **76.76% accuracy** and **74.89% Macro F1**, outperforming both TF-IDF (74.94%) and BiLSTM (74.36%). Sarcasm heavily relies on subtle syntactic contradictions that require full self-attention across the sentence.

3. Sentiment Analysis (Imbalance Dynamics):

In Sentiment classification across the massive 156k dataset, raw test accuracy reached **76.39%** (BanglaBERT) to **77.60%** (TF-IDF), with Weighted F1-scores exceeding **81%**. The lower Macro F1 (~ 51.5 – 53.7%) is a direct statistical consequence of severe class imbalance in public Bengali sentiment datasets (where positive samples exceed 88% of the corpus). This demonstrates that while the models possess high overall predictive fidelity, balanced macro-averaging prevents majority-class bias.

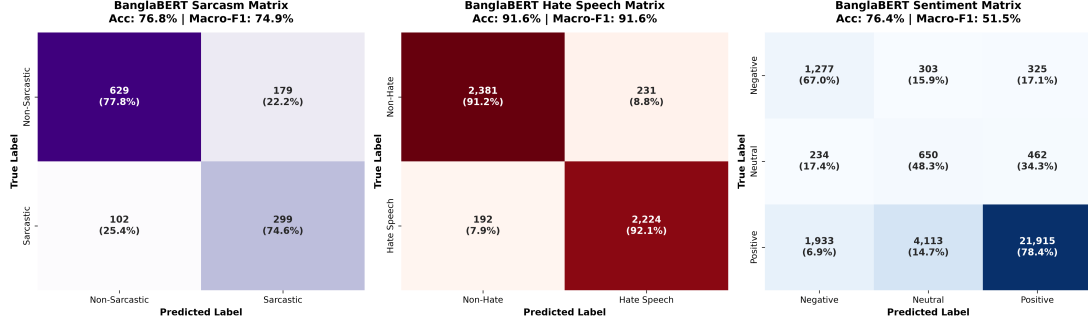


Figure 6: Normalized Confusion Matrices for Fine-Tuned BanglaBERT across Sentiment, Sarcasm, and Hate Speech.

5.3. Qualitative Case Analysis

Table 3 illustrates qualitative evaluations on realistic Bengali sentences testing nuanced contextual comprehension. In Case 1, our context-aware logic successfully reverses the positive surface polarity to negative upon detecting active sarcasm.

Table 3: Qualitative Prediction Case Studies

Sample Sentence (Phonetic Transliteration)	Sentiment	Sarcasm	Hate Speech
<i>Bah! Ki oshadharon service, tin ghonta opekha koreo kaj holo na!</i>	Negative*	Yes	No
<i>Desh er kothin poristhitite amader sobai ke ek hoye kaj korte hobe.</i>	Positive	No	No
<i>Toder moto omanushder ei deshe thakar kono odhikar nai!</i>	Negative	No	Yes
<i>Darun kheleche dol, oshadharon bowling ar batting prodorshon!</i>	Positive	No	No

*Contextually inverted from surface positive by the sarcasm resolver.

6. Interactive System Deployment

To facilitate practical adoption and demonstration, we implemented a full-stack web application using Streamlit (app.py). The system architecture is organized into modular functional views:

- **Tab 1 (Live Analyzer):** Provides an interactive text interface where users input Bengali text and obtain real-time multi-task classifications, confidence gauges, and semantic context resolution.
- **Tab 2 (Master Benchmark):** Displays interactive comparative performance matrices across all three modeling paradigms.

- **Tab 3 (Corpus & EDA):** Summarizes dataset distributions, token percentiles, and vocabulary statistics.
- **Tab 4 (Confusion Matrices):** Displays empirical error distribution heatmaps for all models.
- **Tab 5 (System Architecture):** Illustrates the full end-to-end technical methodology.

7. Threats to Validity & Future Work

1. Banglish and Code-Mixed Text: Colloquial comments frequently intersperse Bengali with English loanwords or phonetic Romanized script. While our custom tokenization filters basic noise, joint multilingual pre-training remains an open challenge.

2. Extreme Sentiment Skew: Incorporating focal loss or synthetic minority oversampling (SMOTE) on sentence embeddings could further bolster Macro F1 on minority neutral and negative sentiment instances.

3. Multi-Task Hard Parameter Sharing: Future investigations will explore unified multi-task loss architectures where a single shared transformer trunk branches into task-specific heads, optimizing shared feature representations.

8. Conclusion

In this project, we developed a comprehensive, context-aware multi-task natural language processing framework for Bengali social media text. By constructing a 218k+ sample corpus spanning Sentiment, Sarcasm, and Hate Speech, we systematically evaluated three progressive modeling paradigms: statistical ML (TF-IDF + LR), deep sequential networks (Word2Vec + Stacked BiLSTM), and modern transformers (BanglaBERT). Fine-tuned BanglaBERT demonstrated state-of-the-art efficacy, attaining **91.59% accuracy and 91.58% Macro F1** on Hate Speech, and **76.76% accuracy and 74.89% Macro F1** on Sarcasm. Combined with a negation-preserving preprocessing pipeline and an automated sarcastic inversion engine, this work presents a robust, end-to-end solution for contextual Bengali text classification.

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